

A Polytropic Approximation of Compressible Flow in Pipes with Friction ¹

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ABSTRACT

This paper demonstrates the usefulness of treating subsonic Fanno flow (adiabatic flow, with friction, of a perfect gas in a constant-area pipe) as a polytropic process. It is shown that the polytropic model allows an explicit equation for mass flow rate to be developed. The concept of the energy transfer ratio is used to develop a close approximation to the polytropic index. Explicit equations for mass flow rate and net expansion factor in terms of upstream properties and pressure ratio are developed for Fanno and isothermal flows. An approximation for choked flow is also presented. The deviation of the results of this polytropic approximation from the values obtained from a traditional gas dynamics analysis of subsonic Fanno flow is quantified and discussed, and a typical design engineering problem is analyzed using the new method.

1. INTRODUCTION

In the design and analysis of industrial systems and equipment, the engineer often encounters the subsonic, compressible flow of gases in pipes, ducts, and other configurations in which friction

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plays a key role. The layout and analysis of process gas distribution systems, the sizing and qualification of safety devices such as pressure relief valves and rupture discs, and the modeling of accident consequences and release rates in the chemical and nuclear industries all require the working engineer to calculate one-dimensional compressible flows with significant friction losses. Subsonic compressible flow is a complex phenomenon, and in order to simplify the required analysis and to allow broadly applicable techniques to be developed, this flow is often modeled as the adiabatic flow of a calorically perfect gas in a constant-area pipe, with friction modeled as a shear stress acting on the fluid from the pipe walls [1]. This model is known as Fanno flow. The theoretical analysis of Fanno flow is well-developed, and practical engineering Fanno flow problems may be solved with the methods of gas dynamics, using implicit equations or tables, typically expressed in terms of critical ratios and Mach numbers [1–6]. Parameters of interest to piping and process engineers, such as pressures and mass flow rates, can then be determined from perfect gas relations.

Since the implicit solution of relatively complex equations is somewhat inconvenient, several explicit, approximate methods are also in widespread use. Since an explicit equation for mass flow rate exists for isothermal flow [4, 7], the flow is sometimes treated as isothermal even when an adiabatic model would be more appropriate. Alternatively, the Darcy-Weisbach formula for incompressible pipe flows, solved in terms of mass flow rate, can be multiplied by a net expansion factor, Y , to match the results of the exact implicit equations. Graphical representations of the net expansion factor are given by Crane [7], and more detailed charts are also available [3], but no explicit equation is given in these references.

By assuming a polytropic relationship between pressure and volume, then applying the concept of the energy transfer ratio [8] to estimate the polytropic index, this paper produces approximate equations for the mass flow rate and net expansion factor for subsonic Fanno flow in terms of the pressure drop $\Delta P/P_1$, friction losses K , and ratio of specific heats γ . These equations are explicit

and fairly simple in form. An approximation for the pressure drop ratio under choked flow conditions is also presented. These approximate explicit equations are compared to those from classical Fanno flow theory, and the results are found to agree to within 1% over the range of non-choked $\Delta P/P_1$, K , and γ typically encountered in engineering practice.

2. TRADITIONAL APPROACH TO FANNO FLOW

The frictional losses of a piping system are represented by the Darcy friction factor f , which represents the number of velocity heads lost along each diameter of pipe length. To account for the frictional and viscous losses in fittings other than straight pipe, a resistance coefficient or K factor is determined for each fitting, and the system K factor is obtained by summing the frictional losses for straight pipe and all fittings: $K = fL/D + \sum_i K_i$. The K factor therefore represents the number of velocity heads lost by the fluid during its flow through the system [7].

Shapiro [1] presented a method for finding the mass flow rate and other parameters of interest once the system frictional losses have been determined. Subsequent references [2, 3, 9] present the same procedure. In brief, the process consists of determining the upstream and downstream Mach numbers of the flow by solving the following simultaneous equations [2]:

$$\frac{P_2}{P_1} = \frac{M_1}{M_2} \left[\frac{2 + (\gamma - 1) M_1^2}{2 + (\gamma - 1) M_2^2} \right]^{\frac{1}{2}} \quad (1)$$

$$K = K^*(M_1) - K^*(M_2) \quad (2)$$

where

$$K^*(M) = \frac{1 - M^2}{\gamma M^2} + \frac{\gamma + 1}{2\gamma} \ln \left[\frac{(\gamma + 1) M^2}{2 + (\gamma - 1) M^2} \right] \quad (3)$$

For subsonic flow in constant-area pipes, the requirement that entropy must increase through the process produces the additional constraint that $M_1 < M_2 < 1$ [2]. Once the upstream and

downstream Mach numbers have been determined, the densities, velocities, and mass flow rate can then be calculated.

Given the complex form of $K^*(M)$, it is not possible in general to solve Eqs. (1) and (2) explicitly. Instead, tabular [2], iterative [5], or graphical [9] techniques must be used. The method of Hullender, Woods, and Huang [10] merits mentioning in that, though it is strictly speaking an iterative method, it produces accurate values for mass flow rate on the second iteration, making it effectively an explicit formula. For routine use in engineering design calculations, however, it is fairly complex to implement in comparison to the common tabular and graphical methods.

3. CALCULATION OF MASS FLOW RATE

An explicit equation for the mass flow rate of Fanno flow can be derived by following a method similar to Shapiro's derivation for the usual Fanno flow equations in terms of Mach number [1], except that the continuity equation is used to express velocity in terms of mass flow rate.

Upstream and downstream control surfaces are placed along the flow path, and mechanical conservation laws applied to the resulting control volume.

For Fanno flow, friction is modeled as a shearing force τ_w , which is proportional to the square of fluid velocity and acts upon the fluid at the pipe wall. Taking this shearing force into account, the momentum balance equation on the control volume is:

$$-AdP - \tau_w dA_w = \dot{m} du \quad (4)$$

Defining the Darcy friction factor, $f \equiv 8\tau_w/\rho u^2$, and hydraulic diameter, $D \equiv 4A \, dL/dA_w$, allows equation (4) to be expressed as

$$-dP - f \frac{\rho u^2}{2} \frac{dL}{D} = \rho u \, du \quad (5)$$

Dividing by the velocity head, $\rho u^2/2$, and rearranging terms produces

$$\frac{2vdP}{u^2} + \frac{2du}{u} + \frac{f}{D}dL = 0 \quad (6)$$

The continuity equation on the control volume is

$$\rho uA = \frac{uA}{v} = \dot{m} = \text{const.} \quad (7)$$

allowing the fluid velocity to be expressed in terms of the mass flow rate and pipe area are constant,

$$du = \frac{\dot{m}}{A}dv \quad (8)$$

Substituting Eq. (7) and Eq. (8) into Eq. (6) produces a combined momentum/continuity equation:

$$\frac{2A^2dP}{\dot{m}^2v} + \frac{2dv}{v} + \frac{f}{D}dL = 0 \quad (9)$$

Since friction is present, the flow is non-isentropic and the relation $Pv^\gamma = \text{const.}$ does not hold. However, for calculational convenience, it is useful here to assume that some polytropic relation $Pv^c = \text{const.}$ may be used, to allow the specific volume to be expressed in terms of the pressure ratio.

$$Pv^c = P_1v_1^c = \text{const.} \quad (10)$$

$$v = \left(\frac{P_1}{P}\right)^{\frac{1}{c}} v_1 \quad (11)$$

The polytropic index c will be determined below through the use of energy considerations. It is worth noting that, as will be shown, c is not a specific constant, but will be permitted to vary with a system's fluid properties, pressure difference, and friction losses.

Substituting into Eq. (9) and integrating over the control volume,

$$\frac{2A^2}{\dot{m}^2 v_1 P_1^{1/c}} \int_{P_1}^{P_2} P^{\frac{1}{c}} dP + 2 \int_{v_1}^{v_2} \frac{dv}{v} + \frac{f}{D} \int_{L_1}^{L_2} dL = 0 \quad (12)$$

$$\frac{2A^2}{\dot{m}^2 v_1 P_1^{1/c}} \left(\frac{c}{c+1} \right) \left(P_2^{\frac{c+1}{c}} - P_1^{\frac{c+1}{c}} \right) + 2 \ln \left(\frac{v_2}{v_1} \right) + K = 0 \quad (13)$$

where $K \equiv fL/D$. Rearranging terms and re-applying Eq. (11) for the specific volume,

$$\frac{2cA^2 P_1}{(c+1) \dot{m}^2 v_1} \left[\left(\frac{P_2}{P_1} \right)^{\frac{c+1}{c}} - 1 \right] + 2 \ln \left[\left(\frac{P_1}{P_2} \right)^{\frac{1}{c}} \right] + K = 0 \quad (14)$$

Solving for $\dot{m}$,

$$\dot{m} = \left[\frac{2c^2 P_1 A^2}{(c+1) v_1} \times \frac{\left(\frac{P_2}{P_1} \right)^{\frac{c+1}{c}} - 1}{2 \ln \left(\frac{P_2}{P_1} \right) - cK} \right]^{\frac{1}{2}} \quad (15)$$

Equations equivalent or similar to Eq. (15) have been presented in chemical engineering texts [11–13], and it is also equivalent to the “corrected conventional adiabatic equation” of Kim and Singh [14], with their n related to the polytropic index by $n = (c - 1)/c$. Also note that, for the case $c = 1$, equation (15) reduces to the complete isothermal equation for pipe flow with friction [7, 14]:

$$\dot{m} = \left[\frac{A^2}{v_1 \left[K + 2 \ln \left(\frac{P_1}{P_2} \right) \right]} \times \frac{P_1^2 - P_2^2}{P_1} \right]^{\frac{1}{2}} \quad (16)$$

4. DETERMINATION OF THE POLYTROPIC INDEX

To determine the polytropic index c assumed in the previous section, the “energy transfer ratio” concept is used [8]. The energy transfer ratio r_{ET} is the instantaneous ratio of heat transfer into a

system to the rate of work performed by the system.

$$r_{ET} = \frac{\delta q}{\delta \mathcal{W}} \quad (17)$$

For a polytropic process, Christians [8] has shown that the polytropic index c is determined by r_{ET} according to the relation

$$c = (1 - \gamma) r_{ET} + \gamma \quad (18)$$

Since Fanno flow is adiabatic, strictly speaking there is no heat transfer in or out of the system and no work performed by or on the system. However, there are external forces operating on the system: specifically, the friction force (modeled as a shear force acting on the fluid at the pipe wall) and the motive force required to move the fluid (modeled as an externally-imposed pressure differential). These external forces will be used to determine the polytropic index.

Neglecting compressibility effects, the friction pressure loss is given by the Darcy-Weisbach equation [7, 15], and the rate of energy dissipated as heat due to friction is the product of this frictional pressure loss and the volumetric flow rate.

$$\dot{q}_f = \Delta P_f u A = K \frac{1}{2} \rho u^2 \times u A = \frac{1}{2} \rho u^3 A K \quad (19)$$

Likewise, some form of work is required to be done, or to have been done, in order to establish and maintain the pressure difference that causes the flow. The minimum rate of work required to establish this pressure difference is that of an isentropic compressor. Marks' Handbook [16] provides an expression for isentropic compressor work, neglecting the kinetic energy of the gas. For steady-state flow, the rate of work required is

$$\dot{\mathcal{W}} = P_2 u_2 A \frac{\gamma}{1 - \gamma} \left[\left(\frac{P_2}{P_1} \right)^{\frac{\gamma-1}{\gamma}} - 1 \right] \quad (20)$$

The work required to produce the flow must cover the flow energy dissipated due to friction as well as the pressure and velocity work. Hence, the expected form of the energy transfer ratio would be

$$r_{ET} = \frac{\delta q}{\delta w} = \frac{\text{friction}}{\text{friction} + \text{work}} = \frac{\dot{q}_f}{\dot{q}_f + \dot{\mathcal{W}}} = \frac{1}{1 + \frac{\dot{\mathcal{W}}}{\dot{q}_f}} \quad (21)$$

Substituting in the expressions for $\dot{q}_f$ and $\dot{\mathcal{W}}$ above produces the following:

$$r_{ET} = \frac{1}{1 + \frac{2P_2}{\rho_2 u_2^2 K} \frac{\gamma}{1-\gamma} \left[\left(\frac{P_2}{P_1} \right)^{\frac{\gamma-1}{\gamma}} - 1 \right]} \quad (22)$$

Notice that the term $u_2^2 \rho_2 / \gamma P_2 = M_2^2$. Because the downstream velocity in a system with a significant pressure drop and significant pressure losses is usually quite high, it is reasonable to make the approximation that $M_2^2 \approx 1$, yielding the following expression:

$$r_{ET} = \frac{1}{1 + \frac{2}{K(1-\gamma)} \left[\left(\frac{P_2}{P_1} \right)^{\frac{\gamma-1}{\gamma}} - 1 \right]} \quad (23)$$

Substituting this energy transfer ratio into Eq. (18) allows the polytropic index to be calculated.

$$c = (1 - \gamma) r_{ET} + \gamma = \frac{(1 - \gamma)}{1 + \frac{2}{K(1-\gamma)} \left[\left(\frac{P_2}{P_1} \right)^{\frac{\gamma-1}{\gamma}} - 1 \right]} + \gamma \quad (24)$$

Since Eq. (24) is somewhat cumbersome, a simplified version may be developed by making a linear approximation.

$$c = 1 + 1.8 \frac{\Delta P}{P_1} \left(\frac{\gamma - 1}{K + 1} \right) \quad (25)$$

A comparison of the predicted values of c using the full Eq. (24) and simplified Eq. (25) is shown

in Figure 1. Although some variation is apparent, testing has shown that using Eq. (15) with c as calculated per Eq. (25) gives a mass flow rate that matches the value obtained from classical Fanno flow theory to within $\pm 1\%$. This is accurate enough to make the simplified equation a useful tool for engineering analysis.

Both the full and simplified equations for c produce values between 1 and γ for the full range of physical values of γ , K , and non-choked $\Delta P/P_1$. The value of c may be thought of as representing where a particular case of adiabatic but non-isentropic flow falls between the extremes of isentropic and isothermal flow. With low friction and high pressure drop, c approaches γ , the isentropic value. Conversely, with high friction or low pressure drop, c approaches the isothermal value of 1.

This also suggests the possibility of modeling systems that fall between adiabatic and isothermal conditions, similar to the “heat transfer ratio” of Hullender et al. [10]. Intermediate conditions can be modeled by manually varying c between the value calculated in Eq. (24) or Eq. (25) and unity. For example, to calculate flow for a system that is midway between adiabatic and isothermal, an effective value of $c_{eff} = 1 + 50\% \times (c - 1)$ can be used.

5. NET EXPANSION FACTOR Y

5.1 USE OF THE COMPRESSIBLE DARCY EQUATION

As mentioned above, the Darcy-Weisbach formula may be used to calculate the pressure drop due to friction of an incompressible fluid flowing in a pipe. In consistent units, the formula for a length of straight pipe is [7, 15]:

$$\Delta P = \frac{1}{2} \rho u^2 f \frac{L}{D} \quad (26)$$

For more complex systems, the total friction loss coefficient for the system, K , may be used as previously described, and the Darcy-Weisbach formula becomes

$$\Delta P = \frac{1}{2}\rho u^2 K \quad (27)$$

Values of K for pipe fittings, valves, and other piping system components are given in [7] or may be provided by the component manufacturer.

For compressible flow, Crane Technical Paper No. 410 [7] gives the “compressible Darcy equation,” produced by applying the net expansion factor, Y , to the incompressible Darcy-Weisbach formula applied to the upstream flow properties, as follows:

$$\Delta P = \frac{1}{2Y^2}\rho_1 u_1^2 K \quad (28)$$

where Y is a function of K and pressure drop $\Delta P/P_1$. Typically Y is determined graphically or by interpolation of tabulated values. Because of its simplicity and since it is recommended in the Crane manual [7], the compressible Darcy equation is widely used in everyday engineering analysis work. It is frequently used in terms of the mass flow rate $\dot{m}$, in the form

$$\dot{m} = AY \sqrt{\frac{2\Delta P \rho_1}{K}} \quad (29)$$

Although, as described above, compressible Fanno flow has an established method of solution through the methods of gas dynamics, the use of the compressible Darcy equation has several advantages when dealing with pipe flow:

1. The familiar Darcy-Weisbach formula can be applied. It is more intuitive to relate a compressible flow problem to a modified version of the pressure drop of the analogous incompressible flow problem, than to switch to the completely different conceptual

framework of Mach numbers and critical properties used in gas dynamics.

2. Pressure, density, and mass flow rate are parameters that can be easily determined with process instrumentation. A gas dynamics analysis of Fanno flow centers on the upstream and downstream Mach numbers, which are not as easily accessible.
3. Often, compressible flows of low pressure drop (and, correspondingly, low Mach number) are treated as incompressible for calculational convenience. Applying Y as a modifying factor allows existing calculations to be easily modified to take compressibility into account, if pressure drop increases or if greater accuracy is desired.
4. When design calculations are performed by hand, by spreadsheet, or with an automated calculation program such as MathCAD, the Y factor can be estimated or interpolated to produce an explicit equation for pressure drop or mass flow rate. Solution using the Mach number typically involves an iterative or numerical approach which may be more cumbersome to implement using these methods.

Accordingly, an explicit and accurate method of calculating the net expansion factor would prove very valuable to practicing design engineers. Since an explicit equation for mass flow rate was developed above, an explicit formula for the net expansion factor can be developed by comparing this expression to the compressible Darcy equation.

5.2 RELATIONSHIP BETWEEN THE POLYTROPIC INDEX AND THE NET EXPANSION FACTOR Y

To develop a formula for the net expansion factor in the compressible Darcy equation, Eq. (29), can be set equal to Eq. (15). Squaring both sides,

$$\frac{2c^2 P_1 A^2}{(c+1) v_1} \times \frac{\left(\frac{P_2}{P_1}\right)^{\frac{c+1}{c}} - 1}{2 \ln\left(\frac{P_2}{P_1}\right) - cK} = A^2 Y^2 \frac{2\Delta P \rho_1}{K} \quad (30)$$

Solving for Y , while noting that $P_2/P_1 = 1 - \Delta P/P_1$, produces an explicit equation for Y in terms of c , K , and $\Delta P/P_1$.

$$Y = \sqrt{\frac{c}{(c+1) \frac{\Delta P}{P_1}} \times \frac{\left(1 - \frac{\Delta P}{P_1}\right)^{\frac{c+1}{c}} - 1}{\frac{2}{cK} \ln\left(1 - \frac{\Delta P}{P_1}\right) - 1}} \quad (31)$$

Also, setting c equal to 1 in Eq. (31) gives an expression for the net expansion factor for isothermal flow.

$$Y_{\text{isothermal}} = \sqrt{\frac{1}{2 \frac{\Delta P}{P_1}} \times \frac{\left(1 - \frac{\Delta P}{P_1}\right)^2 - 1}{\frac{2}{K} \ln\left(1 - \frac{\Delta P}{P_1}\right) - 1}} \quad (32)$$

Since a complete isothermal formula already exists, a Y factor for isothermal flow is not normally calculated; however, having an expression for an isothermal expansion factor allows the use of the same compressible Darcy formula for both adiabatic and isothermal systems.

As described above, the value of c may also be adjusted in order to determine the Y factor for systems that fall between the adiabatic and isothermal extremes.

6. CHOKED FLOW

The method above using Eq. (25) and Eq. (15) or Eq. (31) does not intrinsically account for the choking of the flow – the point where the downstream velocity reaches sonic velocity. Past the choke point, further decreases in downstream pressure cause no further increase in mass flow rate [7]. If the pressure difference is greater than that for choked flow, then the choked pressure difference must be used instead.

One method to create an explicit equation to calculate the choking pressure would be to solve Eq. (3) for M in terms of the choking friction loss factor K^* . This cannot be done exactly, but in the

range $M \in (0, 1)$, $K^*(M)$ can be closely approximated by:

$$K^*(M) \approx \frac{(1 - M^2)(1 - M)}{\gamma M^2} \quad (33)$$

This equation can be solved for M in terms of K^* and γ through the use of the trigonometric cubic formula (see, for example, [17]), and the resulting expression can be used with Eq. (1) to find the critical upstream Mach number. The resulting expressions are rather convoluted but produce values for the choked pressure ratio that are within 2.1% of the theoretical value.

A simpler alternative is to use a best-fit approach. By calculating the critical pressure ratio $\Delta P/P_1$ for a variety of values of γ and K^* , then using the ZunZunSite3 three-dimensional curve fitting software [18] to determine the best fitting curve, the equation below was found to approximate the choked value of $\Delta P/P_1$ over a full range of γ and for $K^* \geq 1$:

$$\left. \frac{\Delta P}{P_1} \right|_{choked} \approx \exp\left(-e^{0.43-0.6\gamma-0.43 \ln K}\right) \quad (34)$$

Other approximations and charts are also available [3, 7, 19]. Alternatively, the upstream Mach number at choked flow may be found in the customary way, by solving Eq. (1) and Eq. (2) numerically for $M_2 = 1$.

7. RESULTS

7.1 ACCURACY OF THE POLYTROPIC APPROXIMATION

The polytropic approximation developed in this paper displays good agreement with the Fanno flow model. The deviations from the theoretical Fanno flow results are due to three simplifying approximations: using the incompressible Darcy formula to compute the energy lost to friction, neglecting the velocity energy in computing the isentropic pumping work requirement, and

approximating the downstream Mach number as one in the resulting expression for r_{ET} .

Counterintuitively, the polytropic approximation remains accurate even in the case of low pressure drop and low Mach numbers. This is partially because, at low pressure drops, the energy transfer ratio is approximately unity and becomes insensitive to errors in the ratio of (assumed) pumping work to friction, or, equivalently, that at low pressure drops the flow approaches isothermal conditions. The errors introduced due to the simplifying assumptions may also cancel out somewhat. The result is that, for a wide range of $\Delta P/P_1$, γ , and K , the level of agreement of the polytropic approximation with Fanno flow theory is better than may be first expected.

Graphical representations of the net expansion factor as calculated using equations (25) and (31) above are shown in Figures 2–4. These figures make clear that the polytropic approximation displays good graphical agreement with theory. Since, by definition, the net expansion factor is a measure of how accurately the mass flow rate is predicted (see Eq. (29)), these plots are also representative of the variance between the theoretical and polytropic calculations of adiabatic mass flow rate.

To obtain an overall idea of the accuracy of the polytropic approximation, iterative and polytropic calculations were compared for a selection of 1,000,000 pseudorandomly-distributed points, in which K was distributed by a power-law distribution in the range $(1, \infty)$, γ was uniformly distributed in $(1, 1.67)$, and $\Delta P/P_1$ was uniformly distributed in $(0, \Delta P/P_1|_{choked})$. The polytropic approximation using the simplified Eq. (25) for c exhibited a standard deviation of 0.18% and a maximum deviation of 0.93%. The maximum deviation appears to occur at choked flow with high values of γ and low values of K . This is when velocity work would be expected to be most significant, and so neglecting it would have the greatest consequence. Higher deviations can occur at values of $K < 1$. Such low values of K , though, are fairly uncommon in engineering practice. Most industrial systems either will have significant frictional losses in the system itself, or will have a full velocity head of frictional loss from discharging into a larger flow area such as

a tank, vessel, or the open atmosphere.

The same pseudorandom numerical test using Eq. (24) to determine c results in a standard deviation of 0.11% and a maximum deviation of 1.0%. For comparison, using the isothermal equation for adiabatic flow results in a maximum deviation of 10.9%, while using the tables for net expansion factor in Crane [7] produces a maximum deviation of 5.9%.

Finally, it is important to note that this polytropic approximation makes no attempt to account for effects that are outside the Fanno flow model, such as intermolecular forces, variable specific heats, or chemical reactions. When such real gas effects are significant, a model that takes these effects into account must be used. Nonetheless, the Fanno flow model has proven itself to be useful across a wide variety of practical engineering applications.

7.2 ACCURACY OF THE CHOKED FLOW APPROXIMATION

The choking pressure as determined by Eq. (34), the best-fit approximation developed in Section 6, is compared to the classical Fanno flow solution in Fig. 5. The approximation shows good visual agreement with the theoretical solution across the full range of γ from 1 to 1.67, and for $K \geq 1$. Note that, as can be seen in the figure, Eq. (34) does *not* accurately predict choking pressure with $K < 1$. As mentioned previously, such low values for K are encountered relatively rarely, but the user must be aware of this limitation.

As in the previous section, calculations were performed, for 1,000,000 pseudorandom points where $1 \leq K \leq \infty$ and $1 \leq \gamma \leq 1.67$, comparing the choking pressure calculated using Eq. (34) to the classical, iteratively calculated Fanno flow solution. The approximation showed a standard deviation of 0.68% and a maximum deviation of 2.1% from the standard method. The largest deviations occurred at low values of γ when $K = 1$.

A modified version of Eq. (34) with an added fourth fitted term was also investigated:

$$\left. \frac{\Delta P}{P_1} \right|_{choked} \approx \exp \left(-e^{0.43-0.6\gamma-0.0005K-0.43 \ln K} \right) \quad (35)$$

The more complex Eq. (35) showed a slightly better visual fit at higher K values, but showed a similar maximum deviation (2.1%) and an only slightly improved standard deviation (0.60%).

8. EXAMPLE

To illustrate the method described in this paper, consider the following example:

PROBLEM

Argon ($\gamma = 1.67$) at 528°R (293 K) and 150 psia (1.034 MPa abs.) static pressure flows through a pressure relief valve and NPS 2 Sch. 40 discharge pipe into a much larger exhaust header maintained at 14.2 psia (97.9 kPa abs.). Frictional losses, including exit loss, can be taken to be $K = fL/D + \sum_i K_i = 31$.

A) What is the mass flow rate of argon into the exhaust header?

B) What is the mass flow rate if the header becomes pressurized to 60 psia (413.7 kPa abs.)?

SOLUTION

Part A

From the problem statement, $\gamma = 1.67$ and $K = 31$. Using Eq. (34),

$$\left. \frac{\Delta P}{P_1} \right|_{choked} = \exp \left(-e^{0.43-0.6\gamma-0.43 \ln K} \right) = 0.8791 \quad (36)$$

which gives a choked downstream pressure of

$$P_2 = P_1 - P_1 \cdot \left. \frac{\Delta P}{P_1} \right|_{choked} = 18.14 \text{ psia (125.1 kPa abs.)} \quad (37)$$

Since this is greater than the downstream pressure in the problem statement, the choked value of P_2 must be used.

The net expansion factor can be found using Eq. (25) and Eq. (31).

$$c = 1 + 1.8 \frac{\Delta P}{P_1} \left(\frac{\gamma - 1}{K + 1} \right) = 1.033 \quad (38)$$

$$Y = \sqrt{\frac{c}{(c + 1) \frac{\Delta P}{P_1}} \times \frac{\left(1 - \frac{\Delta P}{P_1}\right)^{\frac{c+1}{c}} - 1}{\frac{2}{cK} \ln\left(1 - \frac{\Delta P}{P_1}\right) - 1}} = 0.7090 \quad (39)$$

For argon, the density at standard conditions is $\rho_{STP} = 0.1037 \text{ lbm/ft}^3$ (1.661 kg/m^3). Since the upstream temperature is near ambient, the upstream density can be determined using Boyle's Law to be $\rho_1 = P_1/P_{STP} \cdot \rho_{STP} = 1.058 \text{ lbm/ft}^3$ (16.95 kg/m^3). The internal area of an NPS 2 pipe is 3.355 in^2 (2165 mm^2). The compressible Darcy equation (29), with careful attention to unit conversions, now allows the mass flow rate to be calculated.

$$\dot{m} = AY \sqrt{\frac{2\Delta P \rho_1}{K}} = 3.375 \frac{\text{lbm}}{\text{s}} \left(1.531 \frac{\text{kg}}{\text{s}} \right) \quad (40)$$

Part B

Since the downstream pressure now exceeds the choking pressure, the downstream pressure is used as P_2 , and $\Delta P/P_1 = 0.6$. Again, the next expansion factor is found using Eq. (27) and Eq. (33).

$$c = 1 + 1.8 \frac{\Delta P}{P_1} \left(\frac{\gamma - 1}{K + 1} \right) = 1.023 \quad (41)$$

$$Y = \sqrt{\frac{c}{(c + 1) \frac{\Delta P}{P_1}} \times \frac{\left(1 - \frac{\Delta P}{P_1}\right)^{\frac{c+1}{c}} - 1}{\frac{2}{cK} \ln\left(1 - \frac{\Delta P}{P_1}\right) - 1}} = 0.8164 \quad (42)$$

The compressible Darcy equation gives the mass flow rate.

$$\dot{m} = AY \sqrt{\frac{2\Delta P \rho_1}{K}} = 3.211 \frac{\text{lbm}}{\text{s}} \left(1.456 \frac{\text{kg}}{\text{s}} \right) \quad (43)$$

For comparison, an iterative solution to the implicit gas dynamics equations gives mass flow rates of 3.397 lbm/s (1.541 kg/s) for Part A and 3.209 lbm/s (1.455 kg/s) for Part B, differences of 0.66% and 0.046%, respectively.

9. CONCLUSION

By assuming that a polytropic relationship exists for subsonic Fanno flow, an explicit equation for mass flow rate can be obtained. Through the use of Christians' energy transfer ratio concept, a good approximation for the polytropic index can also be obtained explicitly. Taken together, these two explicit formulas provide a good approximation of the mass flow rate and net expansion factor for adiabatic and isothermal flow; previously only implicit, graphical, or tabular solutions were available.

The maximum variance of these approximations, slightly less than 1% from the theoretical value, will usually be less than other uncertainties in the system such as real gas effects or the error in estimating surface roughness, and hence does not represent an impediment to use of this equation for design purposes.

Since the application of Christians' energy transfer ratio to subsonic Fanno and isothermal flows has produced an accurate and useful closed-form approximation, future work may warranted to apply this concept to other common engineering flows, such as supersonic Fanno flow or Rayleigh flow, to determine if similarly useful techniques can be developed.

The polytropic approximation should prove useful in any one-dimensional subsonic flow analysis where the pressure differences are known and the flow rate or exit velocity is to be determined.

An engineer specifying a gas regulator, selecting a rupture disc, or evaluating the release through a crack or hole in a tank now has access to a simple, closed-form method to determine gas flow rate for adiabatic or isothermal flows. Existing graphical, tabular, or numerical techniques remain available; however, the explicit and easily programmable polytropic formulas can be applied in hand calculations or spreadsheets. When calculations are performed with commercial software, the polytropic method provides a quick and straightforward method to independently validate the results. As such, the polytropic approximation represents a very useful tool for the practicing engineer.

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NOMENCLATURE

A	Flow area [length ²]
A_w	Pipe wall area [length ²]
D	Hydraulic diameter [length]
K	Friction loss factor, $K = \frac{fL}{D}$ for a straight pipe [n.d.]
L	Length along pipe or duct [length]
M	Mach number, $M = \frac{u}{\sqrt{\gamma RT}}$ [n.d.]

P	Pressure [force/length ²]
Q	Volumetric flow rate [length ³ /time]
R	Gas constant [energy/mass · temperature]
T	Temperature [temperature]
$\mathcal{W}$	Work [energy]
Y	Net expansion factor [n.d.]
c	Polytropic index, $PV^c = \text{const.}$ [n.d.]
f	Darcy friction factor [n.d.]
$\dot{m}$	Mass flow rate [mass/time]
q	Heat [energy]
r_{ET}	Energy transfer ratio [n.d.]
u	Fluid velocity [length/time]
v	Specific volume [length ³ /mass]

Greek Letters

γ	Ratio of specific heats [n.d.]
ρ	Density, $\rho = \frac{1}{v}$ [mass/length ³]
τ_w	Frictional shearing stress [force/length ²]

Subscripts

1	Upstream conditions
2	Downstream conditions

STP At standard conditions, i.e., atmospheric pressure and room temperature

eff Effective value

f Frictional effects

Superscripts

*

Value of a parameter at sonic (choked) conditions

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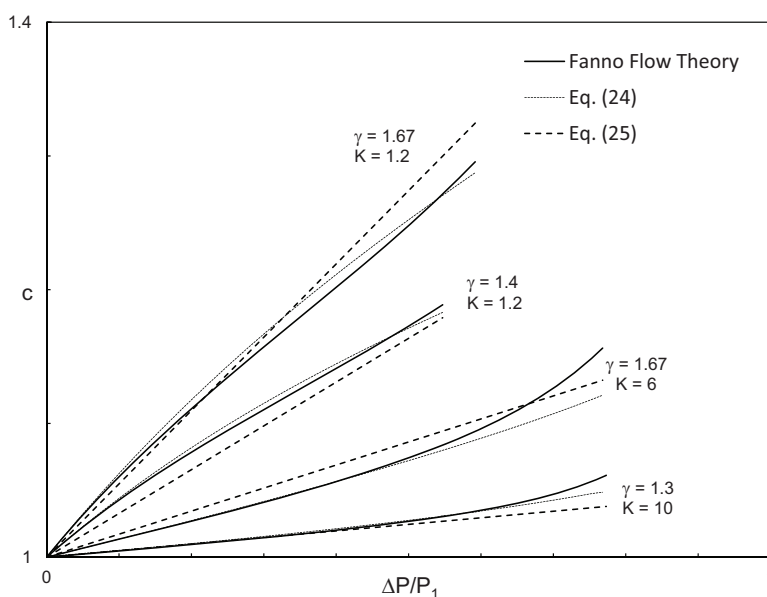


Figure 1. Comparison of Full Eq. (24) and Simplified Linear Eq. (25) for the Polytropic Index c

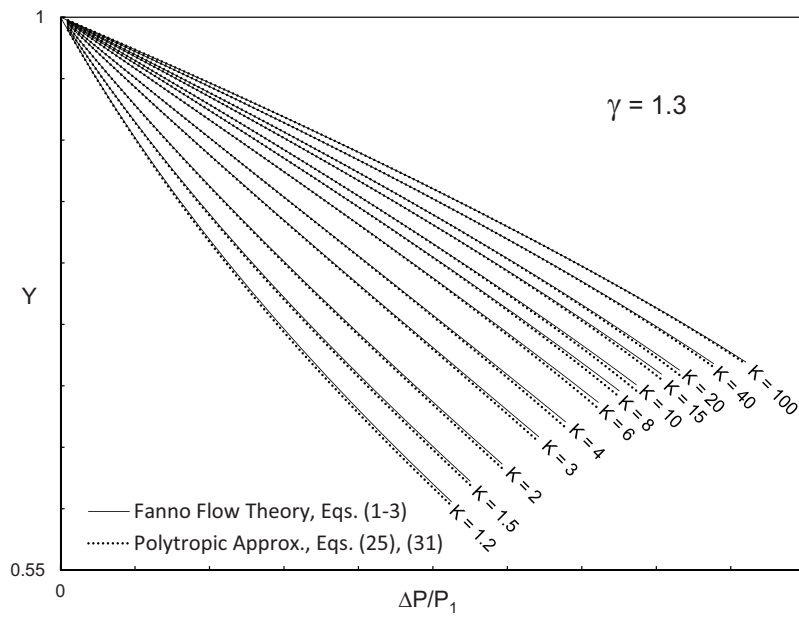


Figure 2. Calculated Net Expansion Factor Y for $\gamma = 1.3$

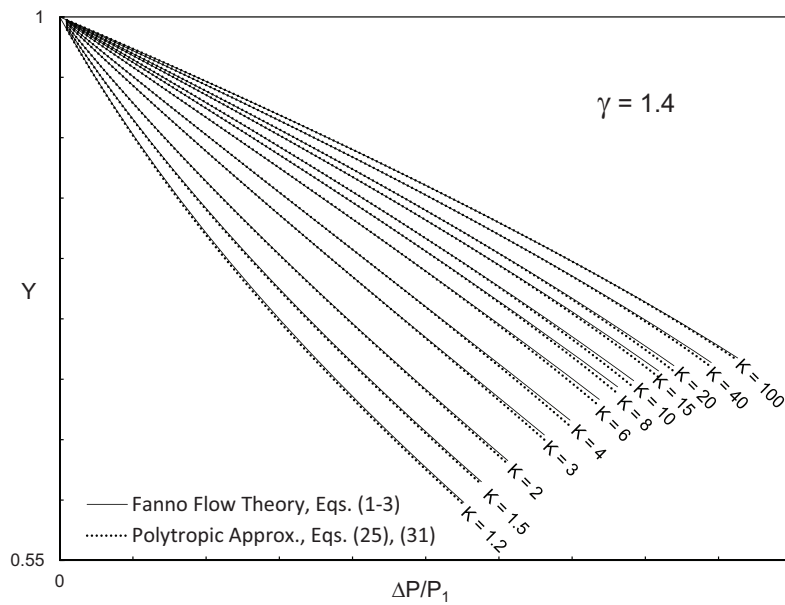


Figure 3. Calculated Net Expansion Factor Y for $\gamma = 1.4$

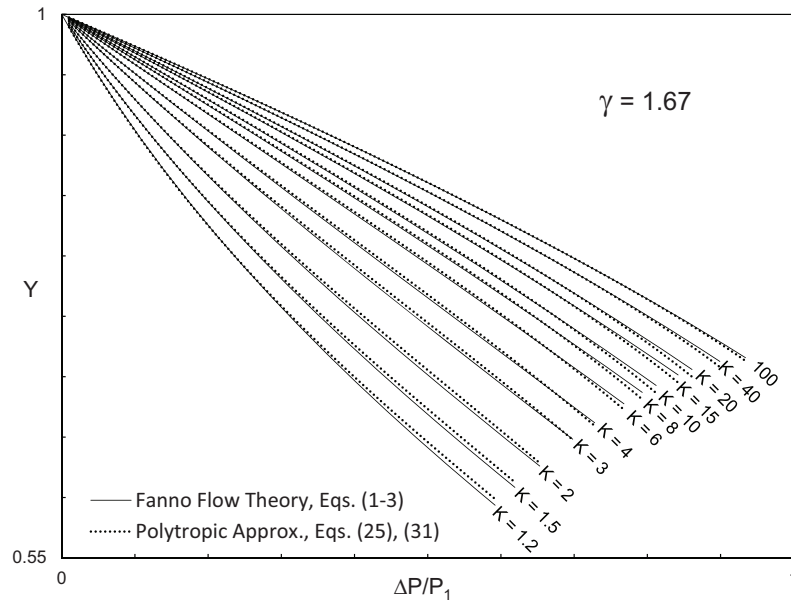


Figure 4. Calculated Net Expansion Factor Y for $\gamma = 1.67$

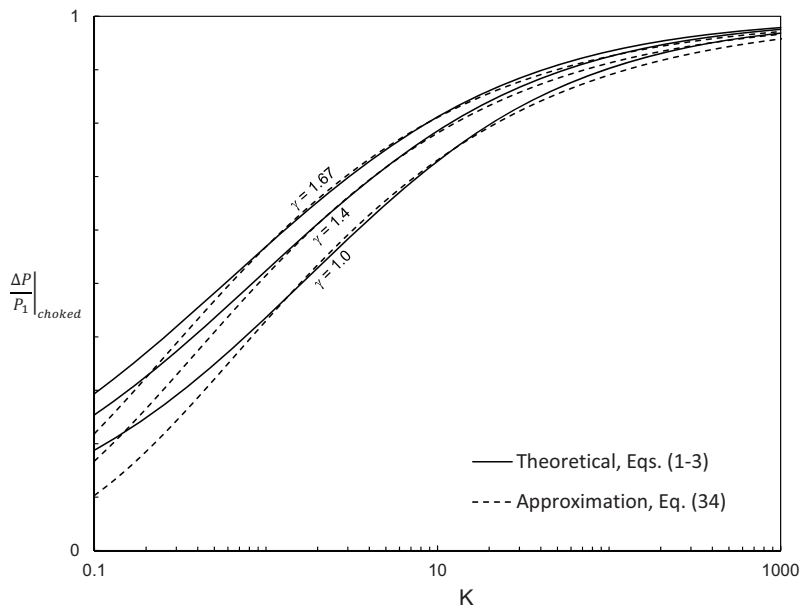


Figure 5. Comparison of Choking Pressure to Best-Fit Approximation, Eq. (34)